JavaScript Events

1. Add an event listener to a button. When we click on it it should display current date and time.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>Event Listener Example</title>
</head>

<body>

<h2>Current Date and Time</h2>

<button id="btn">Show Date & Time</button>

<p id="demo"></p>

<script>

document.getElementById("btn").addEventListener("click", showDate);

function showDate() {
    document.getElementById("demo").innerHTML = new Date();
}

</script>

</body>
</html>
```

2. Show the javascript validation. When you click submit, the text box doesn't need to be empty.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>Form Validation</title>
</head>

<body>

<form onsubmit="return validateForm()">

Name:
<input type="text" id="name">

<input type="submit" value="Submit">

</form>

<script>

function validateForm(){
```

```
let name = document.getElementById("name").value;

if(name == ""){

alert("Name cannot be empty.");

return false;

}

return true;

}

</script>

</body>
</html>
```

3. How console.log() can be used for Debugging.
Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>Console Log Example</title>
</head>

<body>

<script>

let num1 = 20;
let num2 = 30;
let sum = num1 + num2;

console.log("First Number:", num1);
console.log("Second Number:", num2);
console.log("Sum:", sum);

</script>

</body>
</html>
```

How to view the output:

- Open the webpage.
- Press **F12**.
- Open the **Console** tab.
- The values printed using `console.log()` will appear there.

4. Write functions to set a cookie, get a cookie and check a cookie in a single program.
Answer:

```
<!DOCTYPE html>
<html>
```

```html
<head>
    <title>Cookies Example</title>
</head>

<body>

<button onclick="setCookie()">Set Cookie</button>

<button onclick="checkCookie()">Check Cookie</button>

<script>

function setCookie(){

document.cookie = "username=Suman; max-age=86400";

alert("Cookie Set");

}

function getCookie(){

let cookie = document.cookie;

return cookie;

}

function checkCookie(){

let user = getCookie();

if(user != ""){

alert("Cookie Found : " + user);

}

else{

alert("No Cookie Found");

}

}

</script>

</body>
</html>
```

5. Create a JSON object and access it using dot notation.

Answer:

```
<!DOCTYPE html>
<html>
<head>
    <title>JSON Object</title>
</head>

<body>

<script>

let student = {

id:101,
name:"Suman",
course:"BCA",
marks:85

};

document.write("ID : " + student.id + "<br>");

document.write("Name : " + student.name + "<br>");

document.write("Course : " + student.course + "<br>");

document.write("Marks : " + student.marks);

</script>

</body>
</html>
```

Things to know:

1. Debugging in JavaScript.

Answer:

Debugging is the process of finding and fixing errors (bugs) in a JavaScript program.

Common Debugging Methods

- **console.log()** – Prints variable values and messages to the browser console.
- **Browser Developer Tools (F12)** – Inspect code, console output, and errors.
- **debugger keyword** – Pauses code execution for step-by-step debugging.
- **Breakpoints** – Stop execution at specific lines to inspect variables.

Example:

```
let x = 10;
let y = 20;

console.log(x);
console.log(y);
debugger;
console.log(x + y);
```

2. What are Cookies?
Answer:
Cookies are small text files stored in the user's browser. They are used to store information such as user preferences, login details, and session information.

Uses of Cookies

- Store login information.
- Remember user preferences.
- Manage user sessions.
- Track website activity.

Cookie Syntax:

```
document.cookie = "username=Suman";
```

Reading a Cookie:

```
let cookies = document.cookie;
console.log(cookies);
```

Advantages

- Stores small amounts of data on the client.
- Helps maintain user sessions.
- Improves user experience by remembering preferences.